Class 21

Theorem (Kuhn's Theorem)

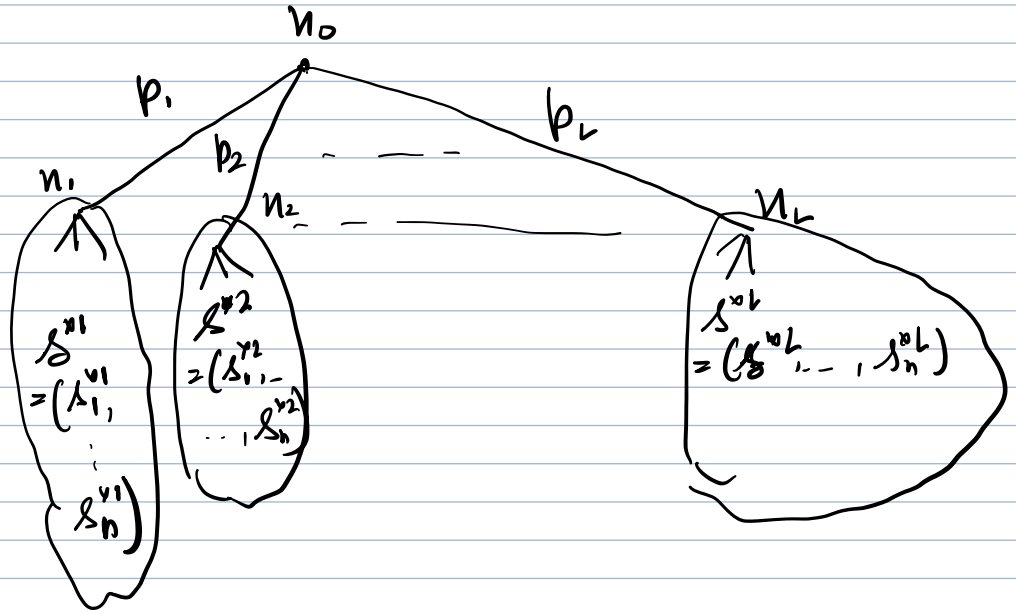
Every extensive form game with perfect information has at least an NE in pure strategies.

Theorem

Every extensive form game with perfect information has at least a SPE in pure strategies

- This equilibrium can be obtained using backward induction principle, i.e., the backward induction equilibrium.

$(\phi, \dots, \phi)$



(i) $n_0 \in V_0$, for player i, consider the strategy s_i^v, which says play s_i^{vl} if the game reaches n_l.

let $s^v = (s_1^v, \ldots, s_n^v)$, then we show that s^v is a SPE.

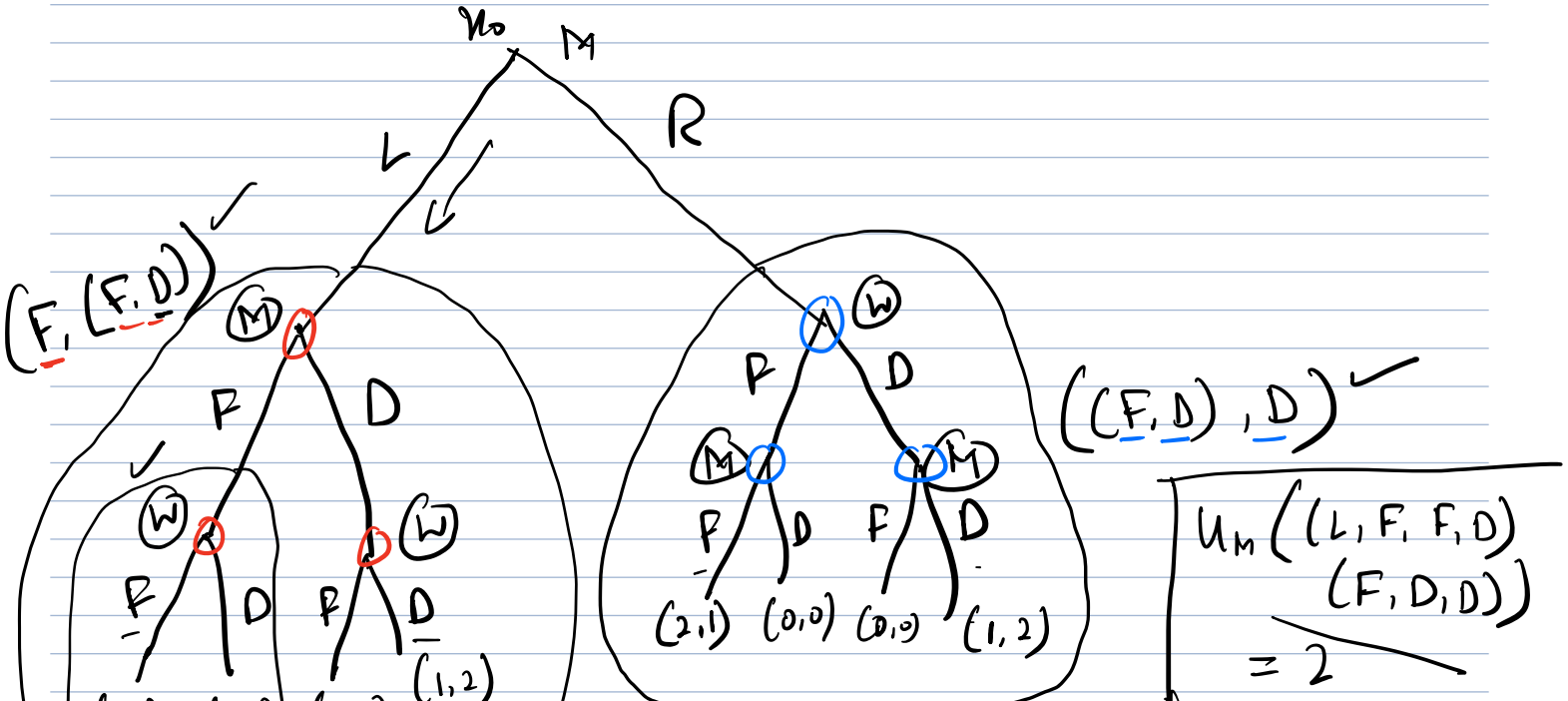
(ii) $n_0 \in V_{i_0}$ $i_0 \in N$

$i \neq i_0$, if the game reaches n_l, play s_i^{vl}
let's call this strategy to be s_i^v

$i = i_0$ then at n_0, choose n_{l_0} if

$$u_{i_0}(s^{vl_0}) \geq u_{i_0}(s^{vl}) \quad \text{for all } l \in \{1, \ldots, L\}$$

and if the game goes to n_l, follow strategy $s_{i_0}^{vl}$.



(2,1) (0,0) (0,0)

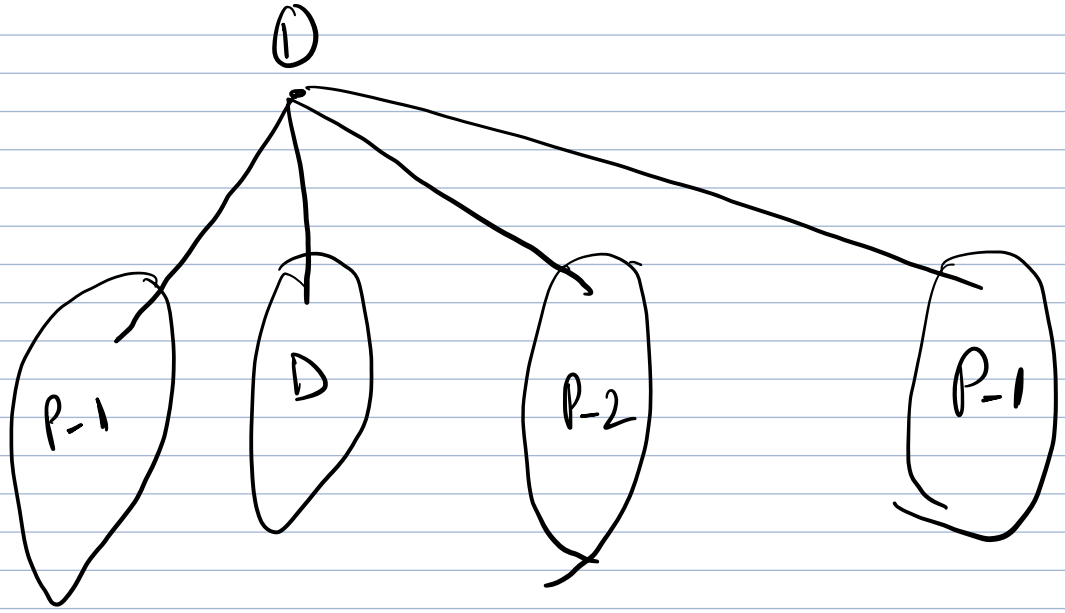
②

①

M - (L, F, (F, D))

W - (F, D, D)

} SPE for the whole game



The exact proof can be found in Solan's book.

	s_2^1 — — —	s_n^2
s_1^1	(1, -1)	
s_2^1	(0, 0)	
s_k^1	(1, -1) — (1, -1) — — — — , (1, -1)	
s_m^1	(-1, 1)	

What is the value of the game chess?

for Chomb $\underline{v} = \bar{v} = 1$